SYLLABUS

MODULE 1 (6 hours)

Entrepreneurship, Business Design, Marketing and Startup Pitching

Entrepreneurial Mindset-SimVenture, GoVenture; Identifying Business Opportunities-Google Trends, SurveyMonkey; Innovation and Business Model Tools- Canvanizer, Miro; Digital Business Models- Shopify, Chargebee; Business Model Canvas-Miro, Lucidchart; Startup Financials and Pitching for Tech Startups-LivePlan, Pitch; Modern Tech Marketing and AI-Driven Decisions- HubSpot, Marketo, Google Analytics, Hotjar, IBM Watson.

MODULE 2 (6 hours)

Introduction to Software Engineering and Agile Software

Introduction to Software Engineering - Professional software development - Git and GitHub/GitLab; Software engineering ethics- OWASP ZAP. Software process models - The waterfall model, Incremental development - Jira, Trello, Lucidchart/ Draw.io. Process activities - Software specification - Figma, UI/UX protyping, Jira; Software design and implementation - Visual Studio Code, Eclipse, Git, GitHub; Software validation-Cypress(web testing), JUnit, Pytest (unit testing), Postman (API testing); Software evolution-VS Code with Git integration, CI tools like CircleCI; Coping with change - Prototyping, Incremental delivery, Boehm's Spiral Model-Adobe XD/Figma, Azure DevOps/GitLab; Agile software development - Agile methods, agile manifesto - values and principles- Scrum, Kanban, XP-Roles like Product Owner, Scrum Master, Dev Team- Agile tools like Jira, Azure Boards, Miro.

MODULE 3 (6 hours)

Agile Development and project management

Agile Management – Agile Software Development-Jira, Azure Boards; Traditional Model vs. Agile Model-Lucidchart, Draw.io, Miro; Classification of Agile Methods-Scrum, Kanban, XP;Agile Manifesto and Principles-Miro, Jamboard; Agile Project Management – Agile Team Interactions-Microsoft Teams, Zoom, Miro, FigJam; Ethics in Agile Teams-Poll Everywhere;Agility in Design-Figma, Adobe XD; Testing-Cypress, PyTest, Postman; Agile Documentations-Google Docs; Agile Drivers, Capabilities and Value-Jira Reports, EazyBI, Google Data Studio, Power BI.

MODULE 4 (8 hours)

Introduction to SCRUM practices in project management

Agile Scrum Framework-Jira Scrum Boards, Azure DevOps; Scrum Artifacts-Jira, Trello; Meetings-Zoom, Microsoft Teams, Miro, FigJam; Activities and Roles-GitHub

Projects; Scrum Team Simulation-ScrumSim, Agile Lego Simulation Kits; Scrum Planning Principles-Planning Poker Online, Miro templates for estimation, Agile Cards; Product and Release Planning-Product Plan(Linear); Sprinting: Planning, Execution, Review and Retrospective-Jira Sprint Boards, Burndown plugins; User story definition and Characteristics- UserStoryMap.io, StoriesOnBoard, Jira User Story Templates, Acceptance tests and Verifying stories-Cucumber, SpecFlow, TestRail, Postman; Burn down chart-Jira Agile Reports, EazyBI, Excel/Google Sheets; Daily scrum- Slack with Geekbot, Microsoft Teams Check-ins; Scrum Case Study-Jira + GitHub, Confluence + Miro, Notion + Trello, LMS platforms like Moodle, Google Classroom.

MODULE 5 (10 hours)

Agile software design and development:

Agile design practices- Figma, Sketch, Adobe XD, Lucidchart, Draw.io, Miro; Role of design Principles; Need and significance of Refactoring-IntelliJ IDEA, VS Code; Refactoring Techniques-ReSharper, Eclipse Refactoring Tools, PyCharm Code Actions; Continuous Integration-GitHub Actions, GitLab CI/CD, CircleCI; Automated build tools-npm, Webpack; Version control-Git, GitHub, GitLab; Agility and Quality Assurance: Agile Interaction Design-Figma, Marvel App, UXPin; Agile approach to Quality Assurance-Zephyr, Allure, Xray for Jira; Test Driven Development-JUnit, NUnit, RSpec, PyTest, Jest; Pair programming: Issues and Challenges-Visual Studio Live Share, CodeTogether; Introduction to Extreme Programming (XP): XP Life cycle-Jira XP Templates, Agilefant; The XP Team, XP Concepts-SonarQube, CodeScene, Refactoring; Technical Debt-SonarCloud, CodeClimate; Timeboxing, Stories, Velocity-Jira Velocity Reports, Burndown Charts, Agile Cards, Scrumpy Poker; Adopting XP: Pre-requisites, Challenges-Atlassian suite (Jira, Bitbucket),Scrum wise, XP-focused Trello boards, Agile Maturity Assessment tools.

Text Books

1. Peter F. Drucker, Innovation and Entrepreneurship, HarperBusiness, 2006.

2. Alexander Osterwalder and Yves Pigneur, Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers, Wiley, 2010.

3. Mike Cohn, Succeeding with Agile: Software Development Using Scrum, Addison-Wesley, 1/e, 2009.

4. David J. Anderson and Eli Schragenheim, Agile Management for Software Engineering: Applying the Theory of Constraints for Business Results, Prentice Hall , 2003.

5. Hazza and Dubinsky, Agile Software Engineering, Series: Undergraduates Topics in Science ,Springer, 2009.

6. Robert C. Martin, Agile Software Development- Principles, Patterns and Practices, Prentice Hall, 2013.

7. Ken Schawber, Mike Beedle, Agile Software Development with Scrum, Pearson, 2001.

MODEL QUESTION PAPER

QP CODE: **Pages: 3**

Reg.No.: ....

Name: ....

MAR ATHANASIUS COLLEGE OF ENGINEERING (AUTONOMOUS), KOTHAMANGALAM

FOURTH SEMESTER B.TECH DEGREE EXAMINATION, MAY 2026

Course Code: B24HU2T03

Course Name: ENTREPRENEURSHIP AND SOFTWARE MANAGEMENT SYSTEMS

Max. Marks: 100 Duration: 3 hours

PART A

Answer all questions. Each question carries 3 marks.

1. Differentiate between Traditional Business Models and Digital Business Models with suitable examples.
2. Explain the importance of Identifying Business Opportunities in entrepreneurship.
3. How does Software Validation help in software development?
4. Why is continuous customer collaboration emphasized in Agile development?
5. Compare the traditional software development model with the Agile model in terms of flexibility, customer involvement, and delivery approach.
6. Explain any three principles of the Agile Manifesto and how they contribute to effective software development.
7. What are the three main Scrum Artifacts, and what is their purpose?
8. Briefly explain the importance of the Daily Scrum meeting in Agile Scrum.
9. What is Refactoring, and why is it important in Agile development?
10. Briefly explain the concept of Test-Driven Development (TDD) and its benefits in Agile.

PART B

Answer any one question from each module. Each question carries 14 marks.

11. (a) Explain the concept of Entrepreneurial Mindset. Discuss how developing an entrepreneurial mindset helps in identifying and evaluating business opportunities in the digital era. (7)

    (b) Analyze the role of Innovation Tools in developing a competitive business model. Illustrate how a startup can use any two innovation tools to improve its business model. (7)

OR

12. (a) Apply the Business Model Canvas framework to develop a business idea for a technology-based startup. Explain each component briefly with respect to your idea. (7)

    (b) Design a marketing strategy for a tech startup using AI-driven decision-making tools. Evaluate how modern tech marketing approaches help startups achieve competitive advantage. (7)

13. (a) Explain the role of Professional Software Development and discuss the key challenges faced by software engineers. (7)

    (b) Compare and contrast the Waterfall Model and Incremental Development in terms of flexibility, risk handling, and efficiency. (7)

OR

14. (a) Describe the key phases of Software Specification, Design, Implementation, Validation, and Evolution with real-world examples. (7)

    (b) Explain how Prototyping and Incremental Delivery help in handling changing customer requirements. (7)

15. (a) Compare and contrast the Traditional Software Development Model with the Agile Model. (7)

    (b) Explain the Agile Manifesto and its core principles with suitable examples. (7)

OR

16. (a) Discuss the significance of Agile Team Interactions and Ethics in Agile Teams. (7)

    (b) How do Agile Drivers, Capabilities, and Value contribute to Agile Project Management? (7)

17. (a) Explain the key roles in the Agile Scrum Framework and their responsibilities. (7)

    (b) Discuss the importance of Sprint Planning, Execution, Review, and Retrospective in Scrum. (7)

OR

18. (a) What is a Burndown Chart? Explain its significance in tracking project progress in Scrum. (7)

    (b) Describe the characteristics of a good User Story and explain how Acceptance Tests help in verifying stories. (7)

19. (a) Explain the role of design principles in Agile Design Practices and their impact on software development. (7)

    (b) Demonstrate how Continuous Integration and Automated Build Tools improve software quality in Agile development with a real-world example. (7)

OR

20. (a) Analyze the challenges of adopting Extreme Programming (XP) and suggest strategies to overcome them. (7)

    (b) Evaluate the effectiveness of Pair Programming in Agile development. Discuss its benefits and potential drawbacks. (7)